Title: Introduction to Business Analysis and AI

Subtitle: Understanding the Intersection of Business Analysis & AI

Lecturer Name: Diptarko Sharma

Date: [07-Apr-25]

Agenda

Introduction to Business Analysis

Introduction to AI

AI in Business

Exercises

A Story

- **Background:**
  Anuj, a BBA graduate from a top-tier Bangalore college, pursued his passion for food by starting a restaurant in Patna. His vision was to provide a seamless dining experience with high-quality, hygienic, and affordable food for middle-class families.
- **Challenges:**
  - After two years, the business was only breaking even, failing to meet his ambitious growth objectives.
  - Lack of differentiation and scalability in a competitive market.
  - Limited customer reach beyond occasional diners.
- **Seeking Solutions:**
  Anuj consults his friend Narayan, a Business Analyst with expertise in problem-solving and strategy. Together, they explore ways to optimize operations, enhance customer engagement, and expand the business towards a sustainable and scalable model.

Understanding Business Analysis

- Understanding business goals
- Using analysis to uncover insights
- Recommending solutions that help the business achieve its objectives.
- It ensures that decision-making is grounded in data and aligned with the strategic direction of the business.

The Story

Background:
Narayan emphasizes the importance of data in identifying problems and solutions.

Data Requirements:
Customer, Sales, Employee, Product, Financial, Geographical, Competitor, Supply Chain, and Social Media Data.

Challenges in Data Availability:
Anuj realizes he only has **Transaction, Product, and Financial Data**. Supply Chain data exists but is unstructured, and social media presence is minimal with limited customer feedback.

Next Steps:
Narayan, undeterred, requests the available data in any format and prepares to conduct an initial analysis, with the possibility of requesting additional data for deeper insights.

Role of a Business Analyst

Key Responsibilities:

- a professional who helps organizations understand their business needs, identify problems
- find solutions to improve processes, products, services, and software through data analysis.
- Who plays a role in aligning the designed and delivered solutions with the needs of stakeholders

Outcome:

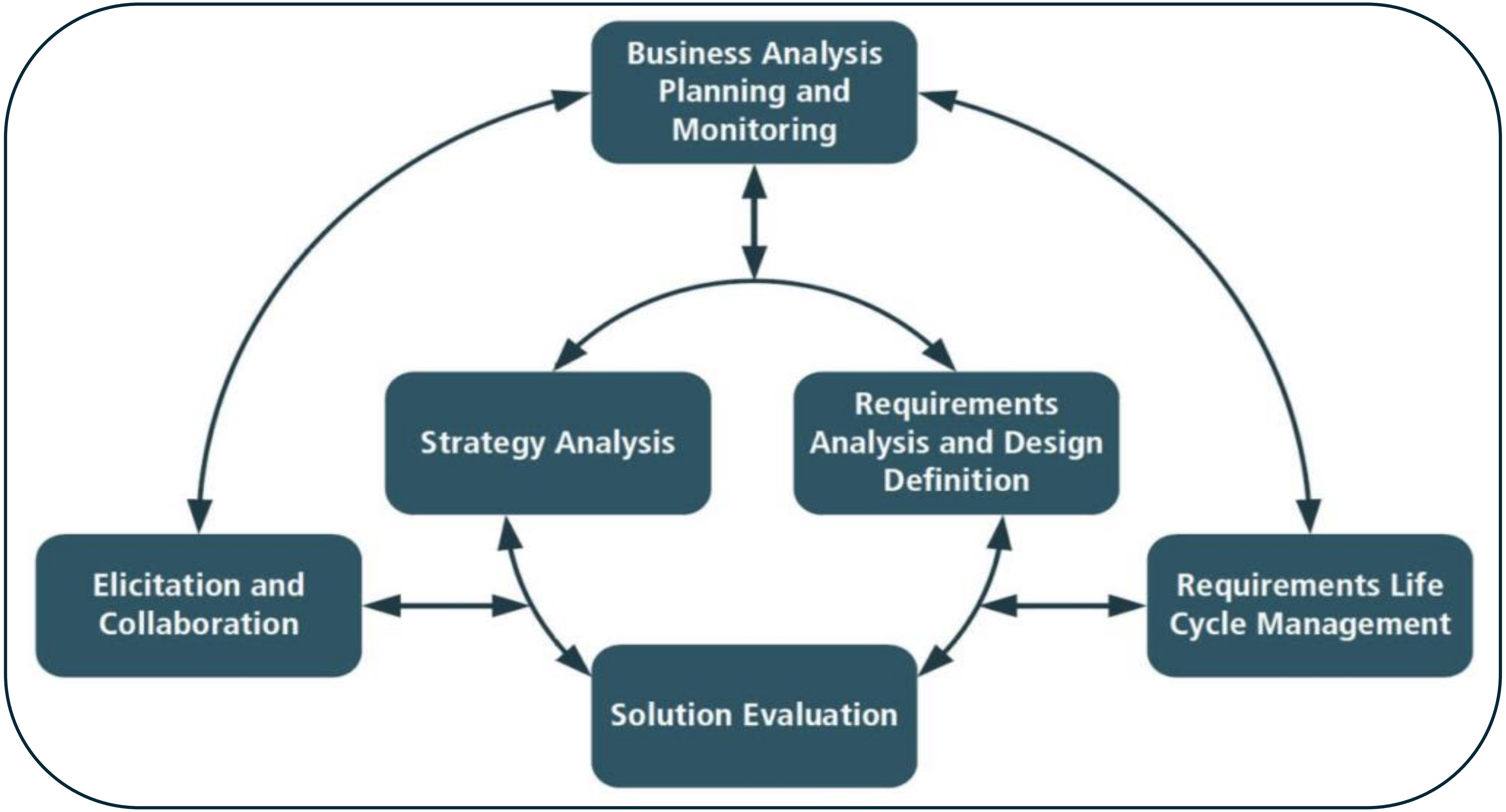
- A Business Analyst ensures successful project delivery and value realization.
- They bridge the gap between IT and the business to help enhance efficiency
- is a key player in organizations, acting as a bridge between business objectives and technological solutions.

BABOK Guide®

BABOK Guide®, which is considered industry standard for Business Analysis, developed by International Institute of Business Analysis (IIBA) says there are six knowledge areas where Business Analysts must focus on:

- **Business Analysis Planning and Monitoring**
- **Elicitation and Collaboration**
- **Requirements Life Cycle Management**
- **Strategy Analysis**
- **Requirements Analysis and Design Definition (RADD)**
- **Solution Evaluation**

BABOK Guide®


Business Analysis Planning and Monitoring
Strategy Analysis
Requirements Analysis and Design Definition
Elicitation and Collaboration
Requirements Life Cycle Management
Solution Evaluation

The Story (Contd…)

Background:

Narayan, a Business Analyst, begins analyzing the data shared by Anuj to identify key business challenges and growth opportunities.

Challenges in Data Analysis:

Many critical datasets like **Customer, Employee, Geographical, and Competitor Data** are missing.

Supply Chain data is **unstructured**, existing only as paper bills.

The available data is **not in a standard format**, making it difficult to derive immediate insights.

Next Steps:

Narayan recognizes the need for data cleaning, structuring, and possible data collection strategies to fill gaps. He plans to extract insights from available data while recommending ways to gather missing information for deeper analysis.

Core Competencies of a Business Analyst

Technical Skills: Business modelling, data analysis, and tools (Microsoft Excel, SQL, BI tools).

Analytical Thinking: Problem-solving with data insights.

Communication Skills: Clear, concise, and effective communication.

Stakeholder Management: Managing diverse expectations and priorities.

Domain Knowledge: understanding of Business Operations, Processes and Strategies

Key Responsibilities

Requirements Gathering	to understand their needs and document these requirements clearly.
Data Analysis	to uncover trends, patterns, and insights that can help make informed business decisions.
Process Improvement	to increase efficiency and effectiveness.
Solution Design	which could include new software applications, process changes, or other innovations, to meet business requirements.
Testing and Validation	proposed solutions are thoroughly tested and validated to meet the business needs effectively.
Communication and Collaboration	Work closely with different teams (IT, management, other stakeholders) to ensure alignment and smooth implementation of solutions.

Impact on Business

Efficiency

Optimizing processes and systems.

Informed Decisions

Providing valuable insights from data analysis.

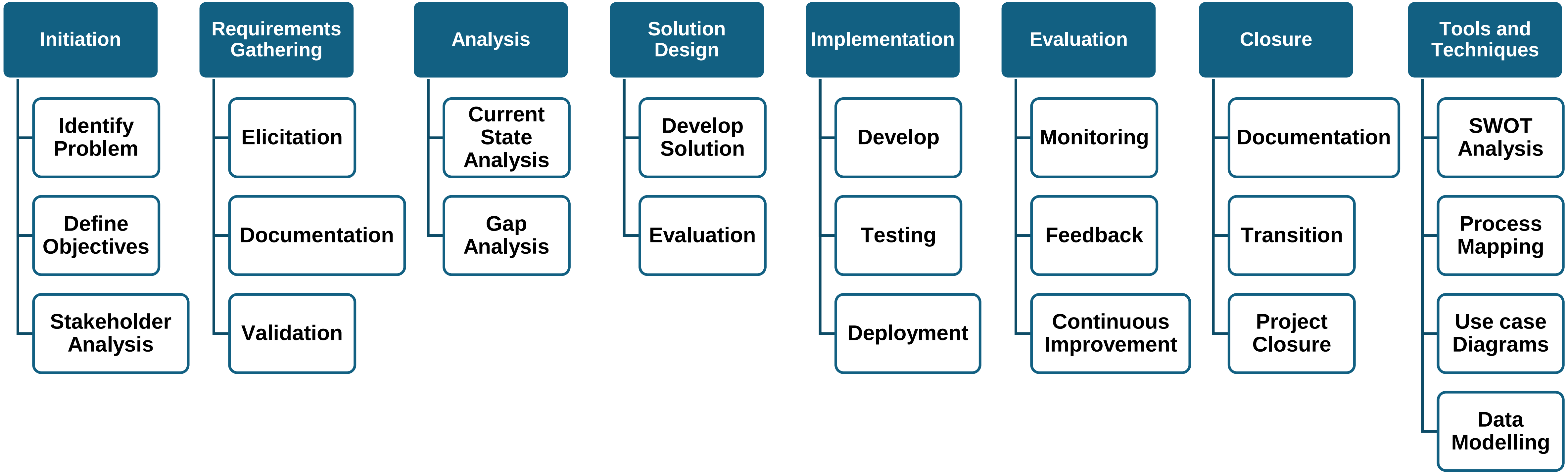
Cost Reduction

Identifying and eliminating inefficiencies.

Stakeholder Satisfaction

Ensuring that business solutions meet the needs of all stakeholders.

Business Analysis Process Overview



Introduction to AI

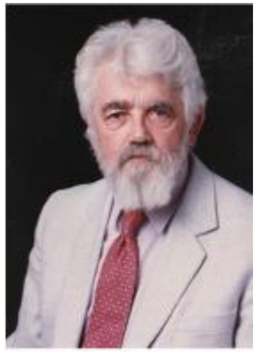
Machines that

mimic human intelligence to perform tasks like

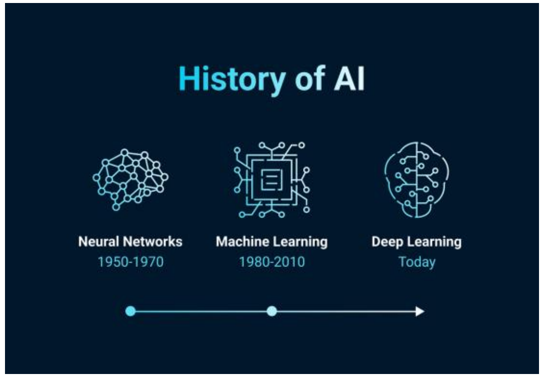
- Decision making
- Speech recognition
- Problem-solving

Founding Fathers of AI

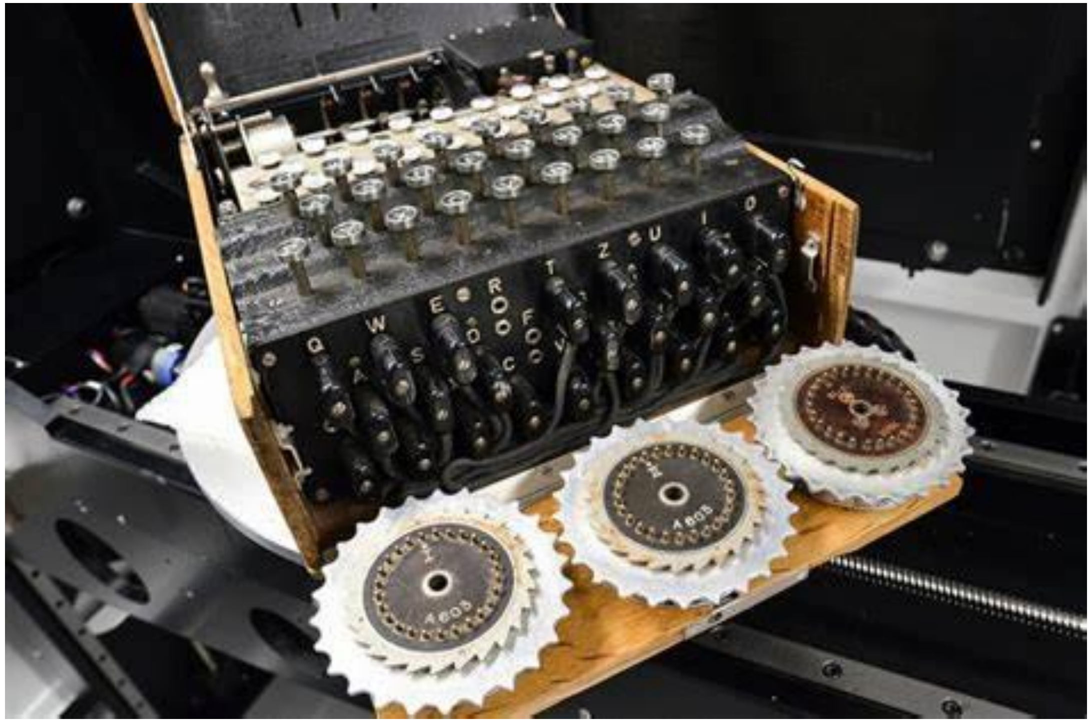
Alan Turing



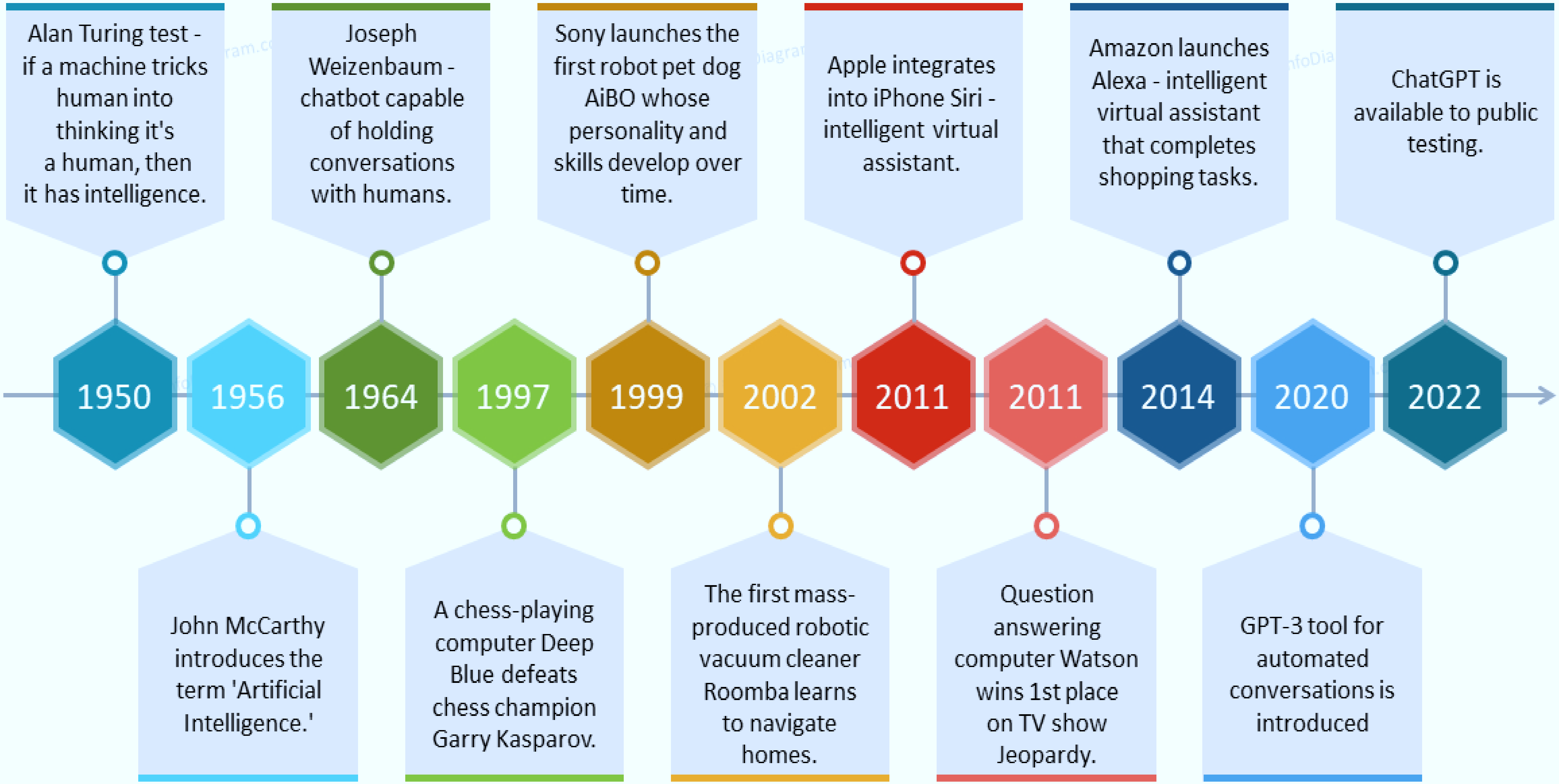
John McCarthy



- **John McCarthy – Father of AI**
  - Coined the term "Artificial Intelligence"
  - Key contributor to AI development
- **Alan Turing – Theoretical Foundation of AI**
  - Featured in *The Imitation Game* movie
  - Played a crucial role in breaking the ENIGMA code
  - Shortened WWII by 2 years, saving millions of lives
  - Introduced the **Turing Test** – a benchmark for AI intelligence
  - Developed the **Turing Machine**, the basis of modern computing



Artificial Intelligence Development History Timeline



AI in

Healthcare
- Diagnostics
- Personalized Medicine
- Robotic Surgery

Finance
- Fraud Detection
- Algorithmic Trading
- Risk Management

Transportation
- Traffic Management
- Route Planning
- Autonomous Vehicle

Retail
- Personalized Recommendations
- Inventory management
- Customer Service

Entertainment
- Content Creation
- Personalized Streaming
- Video games

Education
- Automated Grading
- Virtual Tutors
- Personalized Learning

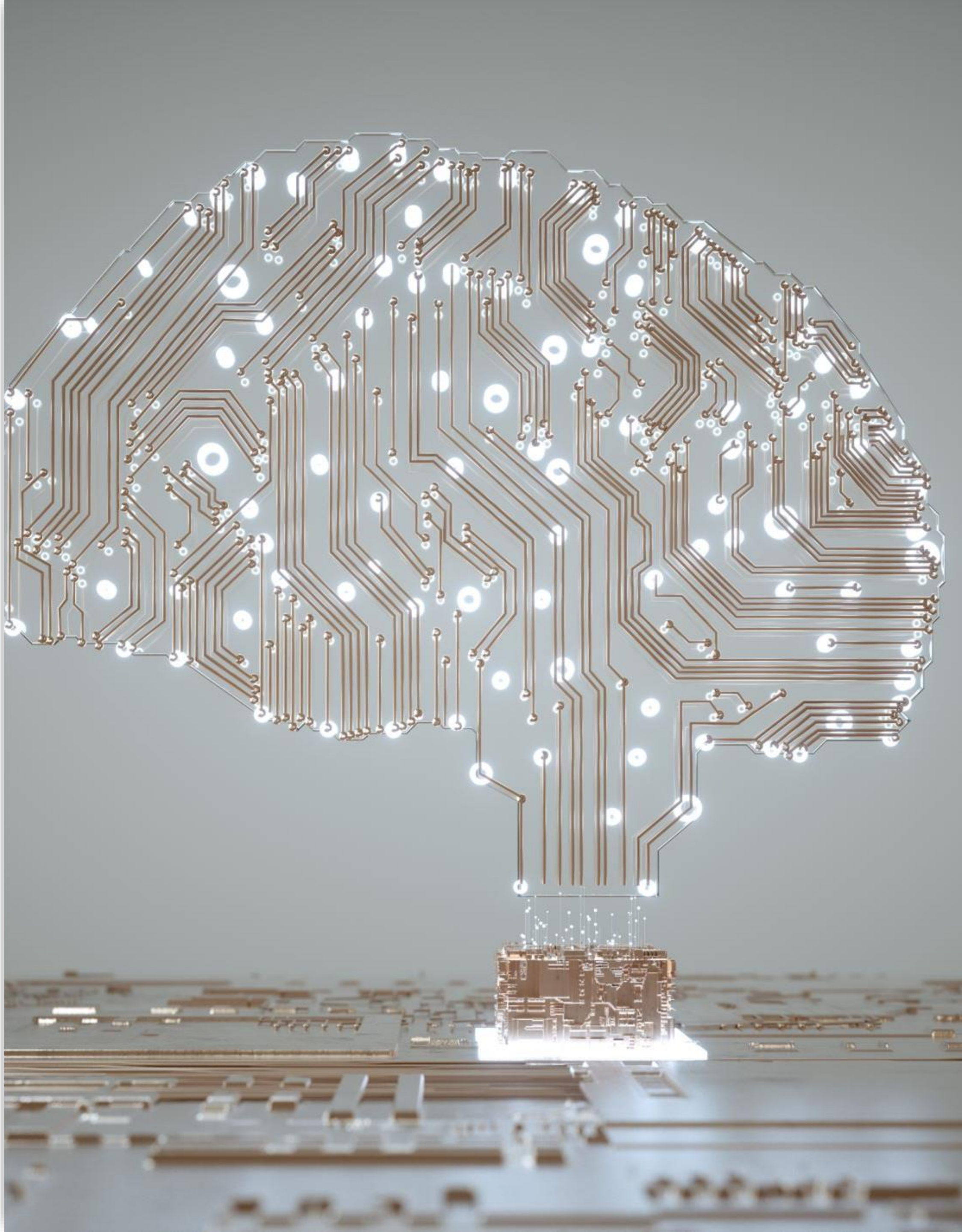
Manufacturing
- Predictive Maintenance
- Quality Control
- Safety

Agriculture
- Crop Monitoring
- Pest Control

Environmental Sustainability
- Renewable Energy
- Wildlife Conservation

Core Areas of AI

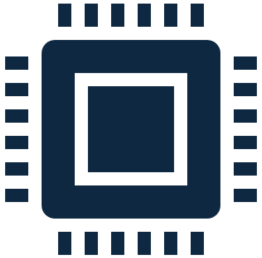
- Machine Learning
- Deep Learning
- Neural Networks
- Natural Language Processing
- Computer Vision
- Reinforcement Learning



Basic Concepts of AI

Artificial Intelligence

Machines mimicking cognitive functions like learning, reasoning, and problem-solving.



Key Concepts:

Algorithms: Instructions for performing tasks.

Data: The raw input required to make decisions.

Computing Power: The infrastructure needed to process data.

How AI Works: 5-Step Process

INPUT

- **Data Collection**: AI collects data from various sources and sorts it into categories.

PROCESSING

- **Data Decoding**: AI analyzes the data using patterns it’s been programmed to recognize, identifying similarities.

OUTCOMES

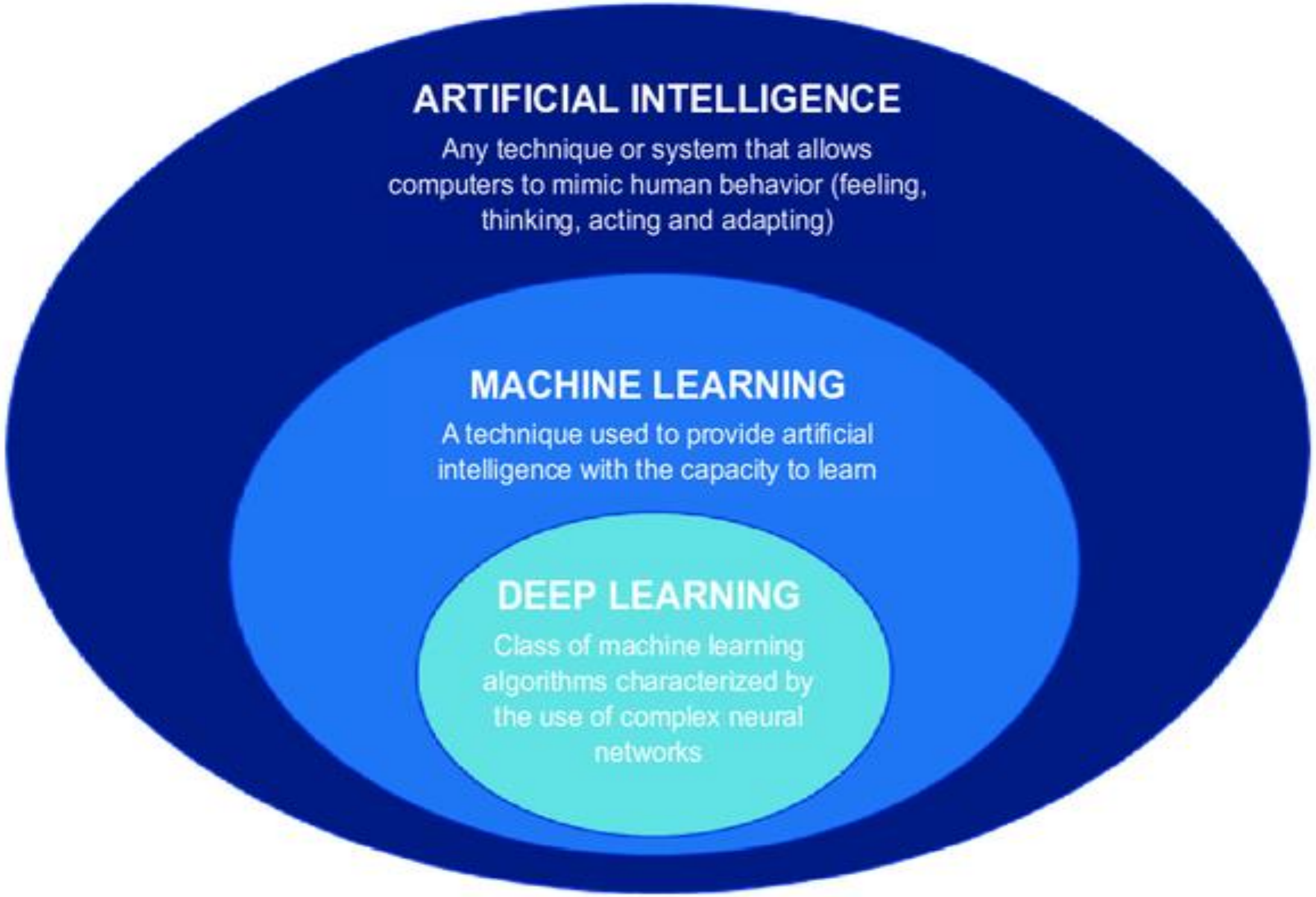
- **Prediction**: Based on recognized patterns, AI predicts potential outcomes.

ADJUSTMENTS

- **Learning from Mistakes**: AI adjusts if outcomes fail, learning from errors and applying changes.

ASSESSMENTS

- **Continuous Improvement**: AI continues to refine its predictions and processes by constantly learning from new data.



[What is Artificial Intelligence? | Artificial Intelligence In 5 Minutes | AI Explained | Simplilearn]

Machine Learning (ML)

ML is a subset of AI that allows systems to learn and improve from experience without being explicitly programmed.

Types of ML:

Supervised Learning: Training models using labeled data.

Unsupervised Learning: Finding hidden patterns in data.

Reinforcement Learning: Learning through trial and error, rewarding correct actions.

Machine Learning: Key Concepts

DATA
- The foundation of ML, crucial for learning.

ALGORITHMS
- Methods for analysing data (e.g., Decision Trees, Neural Networks).

TRAINING
- Process of learning from data to generalize patterns.

FEATURES
- Key attributes influencing model predictions.

LABELS
- Known outcomes in supervised learning.

SUPERVISED LEARNING
- Learning from labelled data (e.g., Classification, Regression).

UNSUPERVISED LEARNING
- Finding patterns in unlabelled data (e.g., Clustering, Dimensionality Reduction).

MODEL EVALUATION
- Measuring performance using accuracy, precision, recall, etc.

OVERFITTING & UNDERFITTING
- Balancing model complexity.

HYPERPARAMETERS
- Tuneable settings controlling model training.

Deep Learning

DL is a subset of ML that uses neural networks with many layers to process large datasets.

Key Uses:

Image Recognition

Speech Recognition

Autonomous Vehicles

Deep Learning: Key Concepts

NEURAL NETWORKS
- The core of deep learning, consisting of multiple layers of neurons.

LAYERS
- Input Layer: Receives raw data.
- Hidden Layers: Extracts patterns and features.
- Output Layer: Produces predictions.

ACTIVATION FUNCTIONS
- Introduce non-linearity (ReLU, Sigmoid, Tanh).

TRAINING
- Adjusts weights using gradient descent to minimize error.

LOSS FUNCTION
- Measures error between predictions and actual values.

BACKPROPAGATION
- Adjusts weights by propagating errors backward.

HYPERPARAMETERS
- Controls learning (learning rate, batch size, layers).

REGULARIZATION
- Prevents overfitting (Dropout, L2 Regularization).

CNNs (Convolutional Neural Networks)
- Specialized for image processing.

RNNs (Recurrent Neural Networks)
- Designed for sequential data (e.g., time series, NLP).

AI in Business Overview

- **AI Impact**: AI has revolutionized industries by optimizing processes, improving decision-making, and creating personalized experiences.

AI in Business: Key Applications

AUTOMATION
- Automates repetitive tasks (e.g., chatbots, data entry).

DATA ANALYSIS
- Processes vast data for insights and predictions.

CUSTOMER SERVICE
- AI chatbots provide 24/7 support.

PERSONALIZATION
- Tailors marketing, products, and services.

SUPPLY CHAIN MANAGEMENT
- Optimizes logistics and inventory.

FRAUD DETECTION
- Identifies unusual transaction patterns.

MARKETING AND SALES
- Enhances campaigns and customer segmentation.

HUMAN RESOURCES
- Streamlines hiring and employee training.

PRODUCT DEVELOPMENT
- Accelerates innovation and market analysis.

DECISION MAKING
- Provides insights for strategic business growth.

AI in Business: HR

Recruitment & Hiring	AI screens resumes and conducts initial interviews.
Employee Onboarding	Personalized AI-driven training programs.
Performance Management	AI analyses productivity and gives feedback.
Employee Engagement & Retention	AI chatbots handle HR queries and predict turnover.
Training & Development	AI recommends learning paths based on career goals.
HR Analytics & Decision Making	AI provides insights for workforce planning.

AI in Business: Marketing

Customer Insights	analyses consumer data to identify trends.
Personalized Marketing	customizes ads, emails, and content for users.
Chatbots & Virtual Assistants	Provide instant customer support and engagement.
Predictive Analytics	forecasts customer behaviour and market trends.
Content Generation	creates automated product descriptions and blog posts.
Ad Optimization	enhances digital ad targeting and bid strategies.
Social Media Analysis	monitors brand sentiment and audience engagement.
Email Marketing	optimizes subject lines, send times, and segmentation.

AI in Business: Supply Chain

Demand Forecasting	predicts demand trends to optimize inventory.
Inventory Management	prevents stockouts and reduces excess inventory.
Logistics Optimization	improves route planning and delivery efficiency.
Supplier Relationship Management	analyzes supplier performance and risks.
Warehouse Automation	AI-powered robots streamline warehouse operations.
Predictive Maintenance	forecasts equipment failures to minimize downtime.
Quality Control	detects defects and ensures product consistency.
Risk Management	identifies potential disruptions in the supply chain.

AI in Business: Banking

Fraud Detection	identifies suspicious transactions and prevents fraud.
Customer Service	chatbots assist customers with banking inquiries 24/7.
Credit Scoring	analyses financial history to assess creditworthiness.
Risk Management	predicts financial risks and market fluctuations.
Personalized Banking	tailors financial products and services for users.
Process Automation	streamlines loan approvals and document processing.
Regulatory Compliance	ensures compliance with banking regulations.
Investment & Wealth Management	provides financial advice and portfolio management.

AI in Business Stock Market

Algorithmic Trading	executes high-speed trades based on market patterns.
Market Prediction	analyses trends to forecast stock prices and movements.
Risk Management	assesses market risks and minimizes potential losses.
Sentiment Analysis	evaluates news and social media to gauge market sentiment.
Portfolio Optimization	helps investors create balanced investment portfolios.
Fraud Detection	identifies suspicious trading activities and insider trading.
Robo-Advisors	provides automated investment advice based on user goals.
High-Frequency Trading	processes massive datasets to execute trades in milliseconds.

MCQ

What is one major role of AI in HR?

A. Replacing human employees entirely

B. Screening resumes and assisting in recruitment

C. Handling payroll manually

D. Reducing employee salaries

Which AI technology is commonly used in Marketing to provide personalized recommendations?

Blockchain

Chatbots

Machine Learning Algorithms

None of the above

In Supply Chain Management, AI helps with:

- Making inventory management more efficient
- Slowing down logistics operations
- Increasing supply chain risks
- Reducing warehouse automation

How does AI improve fraud detection in Banking?

- By allowing transactions without verification
- By detecting unusual transaction patterns
- By manually checking each transaction
- By ignoring risk assessments

What is Algorithmic Trading in the Stock Market?

- Trading stocks using AI-driven strategies
- Buying and selling stocks randomly
- Investing based on human intuition
- A type of fraud in trading

Exercise Overview

Objective: Identify AI solutions for common business challenges.

Approach: Group activity with brainstorming and discussions for each business sector.

AI in HR: Optimizing Employee Recruitment

A global IT company is struggling with an overwhelming number of job applications for various roles. Their HR team manually reviews thousands of resumes, leading to delays, inefficiencies, and potential biases in the hiring process.

Solution

AI- Solution

AI-Powered Resume Screening: AI scans and ranks resumes based on skills and job match.

Chatbot for Candidate Screening: Conducts preliminary interviews and gathers information.

Predictive Hiring Analytics: Identifies candidates most likely to succeed in the company.

Business Impact

Reduced hiring time from months to weeks.

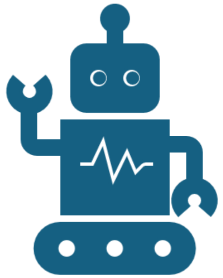
Improved candidate quality with data-driven selection.

Elimination of hiring biases through objective screening.

AI in Marketing – Enhancing Customer Engagement

A fashion e-commerce company is facing low customer engagement and high cart abandonment rates. Despite heavy investments in digital marketing, their campaigns fail to reach the right audience effectively.

Solution



AI- Solution

Personalized Product Recommendations: AI suggests products based on user behavior.

Chatbots for Customer Support: Assists with product queries and order tracking.

AI-Powered Email & Ad Targeting: Optimizes campaigns based on audience insights.

Business Impact

Increased conversion rates due to personalized shopping experiences.

Higher customer retention with targeted recommendations.

Optimized ad spend by reaching the right audience at the right time.

AI in Supply Chain – Reducing Inventory Wastage

A supermarket chain is experiencing significant inventory wastage due to overstocking and poor demand forecasting. Perishable goods often expire before they are sold, leading to financial losses and inefficiencies.

Solution

AI Solution

AI-Driven Demand Forecasting: Predicts demand using sales data and trends.

Automated Inventory Management: Optimizes stock levels to prevent over/under-stocking.

AI-Powered Logistics Optimization: Recommends efficient delivery routes.

Business Impact

Reduced inventory waste by 30-50% through precise demand prediction.

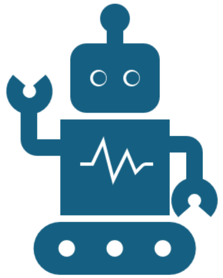
Improved supply chain efficiency with optimized restocking schedules.

Lower operational costs by minimizing excess inventory and spoilage.

AI in Banking – Preventing Fraudulent Transactions

A leading retail bank is struggling with an increasing number of fraudulent transactions, leading to financial losses and declining customer trust. Manual fraud detection is slow and inefficient.

Solution



AI Solution

Real-Time Fraud Detection: Analyzes transaction patterns for suspicious activity.

Behavioral Biometrics: Detects unauthorized access using typing speed and location.

Automated Risk Assessment: Assigns fraud risk scores to transactions for quick action.

Business Impact

Fraud detection accuracy increased by over 90% with AI.

Reduced financial losses through proactive fraud prevention.

Enhanced customer trust with real-time security measures.

Discussion & Summary

Business Analysis:

Identifies business needs and recommends solutions.

Involves data analysis, process improvement, and decision-making.

Helps organizations optimize performance and efficiency.

Artificial Intelligence (AI):

AI automates decision-making and enhances business insights.

Uses Machine Learning (ML) and Deep Learning (DL) for data-driven solutions.

Improves efficiency in customer service, marketing, HR, and supply chain.

AI in Business Analysis:

Automates data collection and trend identification.

Enhances predictive analytics for better decision-making.

Reduces costs and improves business agility.

Growth Strategies for a Mom-and-Pop Store in India

Introduction

Mom-and-pop stores, also known as Kirana stores in India, are small family-owned retail outlets that serve local communities. Despite competition from supermarkets and e-commerce, these stores thrive due to personalized service, credit facilities, and deep customer relationships. This case study explores how **"Gupta General Store"**, a small Kirana shop in Mumbai, improved its operations and profitability using data analysis and digital adoption.

Background

- **Store Name**: Gupta General Store
- **Location**: Mumbai, Maharashtra
- **Owner**: Mr. Ramesh Gupta
- **Established**: 1998
- **Products Sold**: Groceries, snacks, dairy, household items
- **Competitors**: Supermarkets (D-Mart, Reliance Fresh), online grocery (BigBasket, Blinkit)

Challenges Faced

Declining Sales due to competition from e-commerce platforms.

Inventory Mismanagement, leading to overstocking or stockouts.

Lack of Customer Insights, making it hard to offer discounts or loyalty programs.

Limited Digital Presence, missing out on UPI and online orders.

Thank You